1 F2 - Free Group on 2 Generators

1.1 Mathematical Description

The free group $F_2 = \langle a, b \rangle$ is generated by two elements with no relations other than the group axioms. Words reduce to the identity only through free reduction (cancellation of adjacent inverse pairs like `aA` or `Bb`).

1.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse of a (a^{-1})
`b`	Generator b
`B`	Inverse of b (b^{-1})

1.3 Identity Test

A word equals the identity if and only if it freely reduces to the empty word by repeatedly canceling adjacent inverse pairs.

1.4 Dataset Variants

1.4.1 Synthetic (f2_synthetic_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 29.0)

Word	Label	Length
`BBaBbaBAAbaAbB`	0	14
`ABAaAaabAabAABaAAAABBBAbBBBbAb`	0	30
`BAbbabAAaaAAaaBABBab`	1	20
`AABAAbBaabaa`	1	12
`ABbAbbbBabbABaBBaAbbAbaBBAbBBBaBba`	1	34

1.4.2 Augmented (f2_augmented_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 10 – 50 (mean: 29.5)

Word	Label	Length
`BbBBAbBAaABbbbbabaBABaBAAaAbaAAbAa`	0	34
`aBBaaAAbABBABAbbA`	0	17
`bbbaBbabbAaABbaAAaAaaBBABbABBB`	1	30
`AbBbBBbbBbBbBbBbBa`	1	18
`bAbbbBBbAaaAbBbBBBaB`	1	20

1.4.3 Hard (f2_hard_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 20 – 100 (mean: 53.2)

Word	Label	Length
`bbAbAAAAaaAaBABAaBbaabBB`	0	24
`abbababAaAaaAaBBabAaAAaAaBABABBA`	0	32
`aBaABaAbabaaBABAababAABABabBAbaAbA`	1	34
`bbBBbbaAaBBbBbbbBAaABBbbBB`	1	26
`BbBBabbbbbBAaAabAaBBBBBAbbBb`	1	28

2 Z2 - Free Abelian Group on 2 Generators

2.1 Mathematical Description

The free abelian group $\mathbb{Z}^2 = \langle a, b \mid ab = ba \rangle$ is generated by two commuting elements. Words equal the identity when the total count of each generator equals its inverse count.

2.2 Alphabet

Symbol	Meaning
`a`	Generator a
`A`	Inverse of a (a^{-1})
`b`	Generator b
`B`	Inverse of b (b^{-1})

For the anbn variant: only `a` and `b` (lowercase) are used.

2.3 Identity Test

A word equals the identity if and only if:

- $\text{count}(a) = \text{count}(A)$ AND $\text{count}(b) = \text{count}(B)$
- For anbn variant: $\text{count}(a) = \text{count}(b)$

2.4 Dataset Variants

2.4.1 Synthetic (z2_synthetic_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 28.4)

Word	Label	Length
`aBabAAbba`	0	9
`aAABAbBBaBaBABAbAAaBAbaabaB`	0	27
`bBBBbBabbbABbB`	1	14
`aABBAAbbbbBbbaBabBBAAAaaABbaBaaA`	1	32
`BBbBBBbabbBAbb`	1	14

2.4.2 ANBN Variant (z2_anbn_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 31.4)
- **Note:** Uses only lowercase `a` and `b`; identity when count(a) = count(b)

Word	Label	Length
`aabaaaaaaaabbaaaaaaaaa`	0	22
`baabbbaaaaaaaaabbaabbbabababaaa`	0	31
`bbaaba`	1	6
`bbbbaabbaabbaaaabbbbabaaaaab`	1	28
`bbabbabaaaaabbaababbbbababbaaaaabb`	1	34

2.4.3 Hard (z2_hard_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 20 – 100 (mean: 60.1)

Word	Label	Length
`aABBABBbbAabbBabaAbBABAaB`	0	25
`BBBaAaBaAbAaAbbAbABBbAabaBbAAbBbaaA`	0	35
`aAAAAaaABbaAABaAbAaaaaAAaa`	1	26
`AAbbabBababBbbBbBaaBBABbBBbBbAAB`	1	32
`bAbbaBaABbaBbbABAABbABBbaaaB`	1	28

3 B3 - Braid Group on 3 Strands

3.1 Mathematical Description

The braid group B_3 has generators σ_1 and σ_2 with the braid relation: $\sigma_1\sigma_2\sigma_1 = \sigma_2\sigma_1\sigma_2$. This group describes braiding patterns of 3 strands.

3.2 Alphabet

Symbol	Meaning
`s1`	Generator σ_1
`S1`	Inverse σ_1^{-1}
`s2`	Generator σ_2
`S2`	Inverse σ_2^{-1}

3.3 Identity Test

A word equals the identity if it reduces to the trivial braid using the braid relations and free reductions. Computationally verified using braid normal forms.

3.4 Dataset Variants

3.4.1 Synthetic (b3_synthetic_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 26.3)

Word	Label	Length
`S2 S2 S2 s1 s2 S2 S1 s2 s1 s2 S1 s2 s2 s2 S2 S1 s2 S2 S1 s1 s2 S2 s1 S1 S2 s2 s2`	0	27
`s1 S2 s1 S1 s2 s2 s1 S1 S2 s1 S2 s1 S2 s2 s1 s2 S1 s1 S2 S1 s2 S1 s2 s2 S1 S1 S1 S1 s2 S2`	0	30
`S1 s2 s1 s1 S1 s1 s1 s2 S1 S1 S1 S2 s1 S2 s1 s1 s2 S2 S1 S1 s2 S1 s2 s1 s1 s1 S2 S1 S1 s1 S1 S1 S2 s1`	1	34
`s2 S2 s1 S1 s1 S1 S2 s2 s1 S1 S2 s2 S2 s2 S1 s1 S1 s1`	1	18
`s2 S2 s2 S2 S2 s2 s1 S1`	1	8

3.4.2 Augmented (b3_augmented_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 10 – 50 (mean: 29.7)

Word	Label	Length
`s2 s1 S1 s1 S1 S1 s1 S2 s2 S1 s1 s2 s1 s1 s1 s2 S1`	0	17
`s1 S1 S1 s1 S2 s1 S2 s1 s1 S1 s2 S2`	0	12
`S2 s1 S2 s1 s2 S2 s2 S2 s2 S2 s2 S2 S1 s2 S1 s2`	1	16
`s1 s2 S2 s1 S2 S2 S2 s1 s2 s1 s2 S1 s2 S2 s1 S2 S1 S2 S1 s2 s2 s2 S1 s2 S2 S1`	1	26
`S1 S1 s1 s2 s2 s2 S1 s1 s2 S2 S1 S1 s1 s1 S1 s1 S1 s1 S1 S1 s1 s1 s2 S2 S1 s1 S2 S2 S2 S1 s1 s1`	1	32

3.4.3 Hard (b3_hard_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 20 – 80 (mean: 43.4)

Word	Label	Length
s2 s2 S2 S1 s1 S2 s2 S1 s1 S2 s2 S2 S2 s2 s1 s2 S2 s1 S1 s1	0	20
S1 S2 S2 s1 S1 S2 S1 S1 S1 s1 s1 S2 s1 S1 S1 s1 s1 S1 s2 S1 S1 s1 s1 s2 s1 S1 s2 s2 s1	0	29
S2 S1 S2 s1 S2 s1 S1 S1 S2 S2 S2 S1 S1 s1 s1 s2 s2 s2 s1 s1 S1 s2 S1 s2 s1 s2	1	26
s2 s2 S2 S1 S1 S1 S2 s2 s2 s1 S1 S2 S2 s2 s1 s1 s1 s2 S2 S2	1	20
S2 S1 S1 s1 s1 S1 S2 s2 s1 S2 s2 s1 s2 S2 s1 s1 S1 S1 s2 S2 S1 S2 s2 S1 S2 s2 s1 S1 S1 s1 s1 s2	1	32

4 B4 - Braid Group on 4 Strands

4.1 Mathematical Description

The braid group B_4 has generators σ_1, σ_2, σ_3 with relations:

- $\sigma_i\sigma_j = \sigma_j\sigma_i$ when $|i-j| \geq 2$
- $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$ (braid relation)

This group describes braiding patterns of 4 strands.

4.2 Alphabet

Symbol	Meaning
s1	Generator σ_1
S1	Inverse σ_1^{-1}
s2	Generator σ_2
S2	Inverse σ_2^{-1}
s3	Generator σ_3
S3	Inverse σ_3^{-1}

4.3 Identity Test

A word equals the identity if it reduces to the trivial braid using braid relations and free reductions. Computationally verified using braid normal forms.

4.4 Dataset Variants

4.4.1 Synthetic (b4_synthetic_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 26.9)

Word	Label	Length
s3 S1 S1 S1 s2 S1 s1 s2 s2 S1 s2 S1 s2 s1 s1 s1 s2 S2 S3 S1 S2 S1 s1 s2 S3 S2 S2 S2 s2	0	29
s2 S2 S1 s3 S3 S2 s2 s2	0	8
s2 s3 S1 s2 S2 s1 S3 S2 S2 s2 s1 s1 s3 S3 S1 S1 s3 s3 S3 S3 s3 S3 s3 s2 s2 S1 s1 S2 S2 S3	1	30
S3 S3 S2 s2 s3 s3 S3 s2 S1 s1 S2 s3 S3 S3 S3 s3 s3 s3 s3 S1 s1 S3 s2 S2	1	24
s1 s1 S1 S1 s3 S3 s3 S3 s3 S3 s3 S3 s3 S3 S1 s1 S1 s1	1	18

4.4.2 Augmented (b4_augmented_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 10 – 50 (mean: 29.7)

Word	Label	Length
s2 s1 s1 s2 s2 s3 s2 S3 S3 S3 s1 s3 s3 s2 s2 s1 S3 S3 S3 S1 S1 s3 s3 s2 s2 s3 s3 s1 s2 s1 s1 s1 s3 s3	0	34
s2 s3 s3 s2 s2 s2 s1 s2 s1 S1	0	10
s1 s1 S3 S2 S3 s3 S3 s3 S2 s2 s2 s3 S1 S1	1	14
S1 S1 S2 s3 s2 S2 s1 S2 S1 S3 s3 s3 S3 S3 s3 s1 s2 S1 s2 S2 S3 s2 s1 s1	1	24
s1 S3 s3 s2 s2 S3 S2 s1 S2 S3 S3 S2 s2 S2 S1 s1 S3 s3 s2 s3 s3 s2 S1 s2 s3 S2 S2 S3 s3 S1	1	30

4.4.3 Hard (b4_hard_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 20 – 80 (mean: 48.6)

Word	Label	Length
s1 S2 s2 s3 s3 S2 s2 S1 S1 S3 s1 s3 s1 s1 S2 s2 S3 S3 S2 s2 S1	0	21
S3 s3 s1 S1 S2 s2 s3 S3 S2 s2 s2 S2 s2 S2 s2 S2 S1 s1 s1 S1 S3 s3 s1 S1 S2 s2 S2 s2 S2 s2 s2 s3 S3	0	33
S1 S1 s3 s1 s3 S2 s2 S3 S2 s2 S3 S3 s1 s1 s2 S2 S1 S1 s3 s3 S2 s2 s3 S2 s2 S3 S1 S3 s1 s1	1	30
s1 S2 S1 s2 S1 S2 s2 s2 S3 S1 s1 S3 s3 S1 s1 s3 S2 S2 s2 s1 S2 s1 s2 S1	1	24
s3 S2 S3 S3 S2 S3 s3 S1 s1 S2 s2 S1 s1 S2 s2 S1 s1 S3 s3 s2 s3 s3 s2 S3	1	24

5 S4 - Symmetric Group on 4 Elements

5.1 Mathematical Description

The symmetric group S_4 consists of all permutations of 4 elements. Generated by adjacent transpositions with relations:

- $s_i^2 = 1$ (each generator is its own inverse)
- $s_i s_j = s_j s_i$ when $|i - j| \geq 2$
- $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$

5.2 Alphabet

Symbol	Meaning
`s1`	Transposition (1 2)
`s2`	Transposition (2 3)
`s3`	Transposition (3 4)

Note: No inverse symbols needed since $s_i = s_i^{-1}$ in symmetric groups.

5.3 Identity Test

A word equals the identity if the composed permutation is the identity permutation. This is verified by computing the resulting permutation.

5.4 Dataset Variants

5.4.1 Synthetic (s4_synthetic_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 4 – 50 (mean: 26.8)

Word	Label	Length
`s3 s1 s3 s1 s2 s3 s3 s2 s3 s2 s1`	0	11
`s1 s2 s1 s3 s2 s1 s3 s3 s2 s3 s3 s1 s1 s3 s3 s3 s2 s3 s1 s1`	0	20
`s1 s3 s2 s1 s1 s2 s3 s2 s1 s3 s1 s3 s2 s1 s3 s3 s3 s3 s1 s1`	1	20
`s1 s1 s2 s2 s3 s3`	1	6
`s2 s2 s2 s2 s1 s1 s2 s1 s1 s3 s3 s3 s1 s1 s3 s2 s2 s2 s1 s1 s1 s1 s2 s2 s3 s3`	1	26

5.4.2 Augmented (s4_augmented_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 10 – 50 (mean: 29.8)

Word	Label	Length
s2 s2 s2 s1 s2 s3 s2 s2 s2 s2 s1 s3 s2 s1 s2 s3 s1 s1	0	18
s3 s2 s1 s1 s1 s2 s1 s3 s3 s3 s1 s1 s3 s3 s3 s1 s3 s1 s2 s3 s3 s3 s2	0	23
s1 s3 s1 s1 s3 s3 s2 s1 s2 s1 s2 s3 s3 s2 s1 s2 s1 s2 s3 s3 s1 s1 s3 s1	1	24
s1 s3 s3 s2 s1 s1 s1 s3 s3 s1 s3 s1 s1 s3 s3 s3 s1 s3 s3 s1 s1 s1 s2 s3 s3 s1	1	26
s2 s3 s3 s3 s1 s3 s3 s2 s2 s1 s3 s3 s3 s2	1	14

5.4.3 Hard (s4_hard_100k.csv)

- **Rows:** 100,000
- **Label Balance:** 50,000 identity (50.0%) / 50,000 non-identity (50.0%)
- **Length Range:** 20 – 80 (mean: 49.6)

Word	Label	Length
s3 s2 s1 s2 s3 s3 s1 s1 s1 s3 s3 s1 s1 s1 s1 s1 s1 s3 s3 s2 s1 s2 s3	0	23
s2 s2 s3 s1 s1 s3 s2 s1 s1 s3 s3 s1 s1 s1 s1 s1 s3 s3 s2 s3 s1 s1 s3 s2 s2	0	25
s1 s1 s3 s3 s2 s3 s3 s1 s1 s2 s3 s1 s2 s2 s1 s1 s1 s1 s2 s2 s3 s3 s1 s3 s2 s1 s1 s3 s3 s2 s3 s3 s1 s1	1	34
s2 s2 s1 s3 s2 s3 s1 s1 s1 s2 s2 s1 s1 s1 s3 s2 s3 s1 s2 s2	1	20
s1 s2 s2 s2 s2 s1 s1 s3 s1 s3 s1 s3 s3 s3 s3 s1 s1 s1 s1 s3 s1 s3 s1 s1 s2 s2 s2 s2 s1	1	30

6 Summary Statistics

6.1 Dataset Overview

Group	Variant	Rows	Identity %	Non-Identity %	Length Range	Mean Length
F2	Synthetic	100,000	50.0%	50.0%	4 – 50	29.0
F2	Augmented	100,000	50.0%	50.0%	10 – 50	29.5
F2	Hard	100,000	50.0%	50.0%	20 – 100	53.2
Z2	Synthetic	100,000	50.0%	50.0%	4 – 50	28.4
Z2	ANBN	100,000	50.0%	50.0%	4 – 50	31.4
Z2	Hard	100,000	50.0%	50.0%	20 – 100	60.1
B3	Synthetic	100,000	50.0%	50.0%	4 – 50	26.3
B3	Augmented	100,000	50.0%	50.0%	10 – 50	29.7
B3	Hard	100,000	50.0%	50.0%	20 – 80	43.4
B4	Synthetic	100,000	50.0%	50.0%	4 – 50	26.9
B4	Augmented	100,000	50.0%	50.0%	10 – 50	29.7
B4	Hard	100,000	50.0%	50.0%	20 – 80	48.6
S4	Synthetic	100,000	50.0%	50.0%	4 – 50	26.8
S4	Augmented	100,000	50.0%	50.0%	10 – 50	29.8

Group	Variant	Rows	Identity %	Non-Identity %	Length Range	Mean Length
S4	Hard	100,000	50.0%	50.0%	20 – 80	49.6

6.2 Totals by Group

Group	Total Examples	Dataset Files
F2	300,000	3
Z2	300,000	3
B3	300,000	3
B4	300,000	3
S4	300,000	3
Total	**1,500,000**	**15**